WW-C-440b
October 16, 1965
SUPERSEDING
Fed. Spec. WW-C-440a
August 25, 1958

FEDERAL SPECIFICATION

CLAMPS, HOSE, (LOW-PRESSURE)

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers metal clamps for hose (see 6.1).

1.2 Classification.

1.2.1 *Types.* Clamps shall be of the following types as shown in figure 1 (see 6.2).

A—Wire clamp with tangential-attached machine screw adjusting mechanism.

B—Band clamp with circumferential slots and tangential-attached machine screw adjusting mechaniam.

C—Band clamp with radial-attached machine screw or thumb screw adjusting mechanism.

D—Band clamp with tangential-attached machine screw adjusting mechanism.

E—Spring clamp, self-tightening.

F—Band clamp with worm-type tangential-attached, slotted or thumb screw adjusting mechanism.

G—Band clamp with tangential-attached machine screw adjusting mechanism (see 6.5).

H—Band clamp, with nonadjusting lock.

2. APPLICABLE SPECIFICATIONS, STANDARDS, AND OTHER PUBLICATIONS

2.1 Specifications and standards. The following specifications and standards, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

Federal Specifications:

QQ-B-613—Brass, Leaded and Non-Leaded: Flat Products (Plate, Bar, Sheet and Strip).

QQ-B-626—Brass, Leaded and Non-Leaded; Rod, Shapes, Forgings and Flat Products with Finished Edges (Bar, Flat Wire and Strip).

QQ-P-35—Passivation Treatment for Austenitic, Ferritic, and Martensitic Corrosion-Resisting Steel (Fastening Devices).

QQ-P-416—Plating, Cadmium (Electrodeposited).

QQ-Z-325—Zinc Coating, Electrodeposited, Requirements for.

Federal Standard:

Fed. Std. No. 66—Steel: Chemical Composition and Hardenability.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington, D. C. 20402.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are

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available without charge at the General Services Administration Regional Offices in Boston, New York, Washington, D. C., Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, and Seattle, Washington.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Specifications:

MIL-H-775—Hose, Rubber or Fabric (Including Tubing), and Fittings, Nozzles and Strainers, Packaging of.
MIL-H-6000—Hose, Rubber (Fuel, Oil, Coolant, Water and Alcohol).
MIL-L-6082—Lubricating Oil, Aircraft Reciprocating Engine (Piston).

Military Standard:

MIL-STD-105—Sampling Procedures and Tables for Inspection by Attributes.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 **Other publications.** The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids, or request for proposal shall apply.

Society of Automotive Engineers Publication:

SAE Handbook.

(Application for copies should be addressed to the Society of Automotive Engineers, Inc., 485 Lexington Ave., New York, N. Y., 10017.)

National Bureau of Standards Handbook:

H28—Part—Screw-Thread Standards for Federal Services.

(Application for copies should be addressed to the Superintendent of Documents, U. S. Government Printing Office, Washington, D. C., 20402.)

3. REQUIREMENTS

3.1 **Material.**

3.1.1 *Carbon steel.* Unless otherwise specified (see 6.2), material for clamps or components shall be carbon steel. Material shall conform to C1010 thru C1022 of Fed. Std. No. 66.

3.1.2 *Corrosion-resisting steel.* When specified (see 6.2), material for clamps and components shall be corrosion-resisting steel. Material shall conform to steel numbers 201, 202, 301 thru 304, 410, 416, 430, 430F, or 431 of Fed. Std. No. 66.

3.1.3 *Spring steel.* When type E clamps are specified (see 6.2), they shall be made of spring steel. Material shall conform to steel numbers C1065 through C1080 of Fed. Std. No. 66.

3.1.4 *Brass.* When specified (see 6.2), clamps or components shall be made of brass. Material shall conform to composition 11 of QQ-B-613 or composition 22 of QQ-B-626.

3.2 **Design and construction.**

3.2.1 *Design.* Clamps shall be so designed that all component parts of the clamp shall remain as an assembly during installation and removal of the clamps. Type designations of 1.2.1 are in accordance with SAE Handbook.

3.2.2 *Construction.* The clamps shall hold the hose substantially 360° around the circumference of the hose without leakage at the preasures specified in 3.4.5.

3.2.2.1 *Size of clamps.* Clamp size shall be designated by the outside diameter of the hose. Unless otherwise specified (see 6.2), clamps shall be furnished in the sizes specified in table I. The A-diameters in table IA refer to the inside diameter of the clamp.

3.2.3 *Threads.* Threads for the clamp adjusting mechanism shall be Unified and American Standard, classes 2A and 2B for screws and nuts, respectively, in accordance with Handbook H28. Threads for type F clamps shall be a modified buttress thread type, standard with clamp manufacturers.

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TABLE I. *Clamp types available for listed hose diameters (Size of clamp is the same as "OD" of hose; (X) indicates types available)*

Nominal clamp size	Clamp types								Nominal clamp size	Clamp types							
	A	B	C	D	E	F	G	H		A	B	C	D	E	F*	G	H
3/8	--	--	--	--	X	--		X	4-9/8	X	--	X	X	--	--	X	--
7/16	X	--	--	--	X	--		X	4-11/16	X	--	X	X	--	--	X	--
1/2	X	X	--	--	X	--		X	4-3/4	X	--	X	X	--	X	X	--
9/16	X	X	--	--	X	--		X	4-13/16	X	--	X	X	--	--	X	--
5/8	X	X	--	X	X	--		X	4-7/8	X	--	X	X	--	--	X	--
11/16	X	X	--	X	X	--		X	4-15/16	X	--	X	X	--	--	X	--
3/4	X	X	--	X	X	--	X	X	5	--	--	X	X	--	X	X	X
25/32	--	--	--	--	--	X	X	--	5-1/16	--	--	--	X	--	--	X	--
13/16	X	X	--	X	X	--	X	X	5-1/8	--	--	--	X	--	--	X	--
7/8	X	X	--	X	X	--	X	X	5-3/16	--	--	--	X	--	--	X	--
29/32	--	--	--	--	--	X	X	--	5-1/4	--	--	--	X	--	X	X	--
15/16	X	X	X	X	X	--	X	X	5-5/16	--	--	--	X	--	--	X	--
1	X	--	X	X	X	X	X	X	5-3/8	--	--	--	X	--	--	X	--
1-1/32	--	X	--	--	--	--	X	--	5-7/16	--	--	--	X	--	--	X	--
1-1/16	X	X	X	X	X	X	X	X	5-1/2	--	--	--	X	--	X	X	--
1-1/8	X	X	X	X	X	--	X	X	5-9/16	--	--	--	X	--	--	X	--
1-3/16	X	X	X	X	X	--	X	X	5-5/8	--	--	--	X	--	--	X	--
1-1/4	X	X	X	X	X	X	X	X	5-11/16	--	--	--	X	--	--	X	--
1-5/16	X	X	X	X	X	--	X	X	5-3/4	--	--	--	X	--	X	X	--
1-3/8	X	X	X	X	X	X	X	X	5-13/16	--	--	--	X	--	--	X	--
1-7/16	X	X	X	X	--	--	X	X	5-7/8	--	--	--	X	--	--	X	--
1-1/2	X	X	X	X	X	X	X	X	5-15/16	--	--	--	X	--	--	X	--
1-9/16	X	X	X	X	--	--	X	X	6	--	--	--	X	--	X	X	X
1-5/8	X	X	X	X	X	--	X	X	6-1/4	--	--	--	--	--	X	X	--
1-11/16	X	X	X	X	--	--	X	X	6-1/2	--	--	--	--	--	X	X	--
1-3/4	X	X	X	X	X	X	X	X	6-3/4	--	--	--	--	--	X	--	--
1-13/16	X	X	X	X	--	--	X	X	7	--	--	--	--	--	X	--	--
1-7/8	X	X	X	X	X	--	X	X	7-1/4	--	--	--	--	--	X	--	--
1-15/16	X	X	X	X	X	X	X	X	7-1/2	--	--	--	--	--	X	--	--
2	X	X	X	X	X	X	X	X	7-3/4	--	--	--	--	--	X	--	--
2-1/32	--	X	--	--	--	--	X	--	8	--	--	--	--	--	X	--	--
2-1/16	X	--	X	X	--	--	X	--	8-1/4	--	--	--	--	--	X	--	--
2-1/8	X	X	X	X	X	--	X	X	8-1/2	--	--	--	--	--	X	--	--
2-3/16	X	X	X	X	X	--	X	X	8-3/4	--	--	--	--	--	X	--	--
2-1/4	X	X	X	X	X	X	X	X	9	--	--	--	--	--	X	--	--
2-9/32	--	X	--	--	--	--	X	--	9-1/4	--	--	--	--	--	X	--	--
2-5/16	X	--	X	X	--	--	X	X	9-1/2	--	--	--	--	--	X	--	--
2-3/8	X	X	X	X	X	--	X	X	9-3/4	--	--	--	--	--	X	--	--
2-7/16	X	--	X	X	--	--	X	--	10	--	--	--	--	--	X	--	--
2-1/2	X	X	X	X	X	X	X	X	10-1/4	--	--	--	--	--	X	--	--
2-9/16	X	--	X	X	--	--	X	X	10-1/2	--	--	--	--	--	X	--	--
2-5/8	X	X	X	X	X	--	X	X	10-3/4	--	--	--	--	--	X	--	--
2-11/16	X	--	X	X	--	--	X	--	11	--	--	--	--	--	X	--	--
2-3/4	X	X	X	X	--	X	X	X	11-1/4	--	--	--	--	--	X	--	--
2-13/16	X	--	X	X	--	--	X	X	11-1/2	--	--	--	--	--	X	--	--
2-7/8	X	X	X	X	--	--	X	X	11-3/4	--	--	--	--	--	X	--	--
2-15/16	X	--	X	X	--	X	X	X	12	--	--	--	--	--	X	--	--
3	X	X	X	X	--	X	X	X	12-1/4	--	--	--	--	--	X	--	--
3-1/16	X	--	X	X	--	--	X	X	12-1/2	--	--	--	--	--	X	--	--
3-1/8	X	X	X	X	--	--	X	X	12-3/4	--	--	--	--	--	X	--	--
3-3/16	X	--	X	X	--	--	X	--	13	--	--	--	--	--	X	--	--
3-1/4	X	X	X	X	--	X	X	--	13-1/4	--	--	--	--	--	X	--	--
3-5/16	X	--	X	X	--	--	X	X	13-1/2	--	--	--	--	--	X	--	--
3-3/8	X	X	X	X	--	--	X	X	13-3/4	--	--	--	--	--	X	--	--
3-7/16	X	--	X	X	--	--	X	--	14	--	--	--	--	--	X	--	--
3-1/2	X	--	X	X	--	X	X	X	14-1/4	--	--	--	--	--	X	--	--
3-9/16	X	X	X	X	--	--	X	X	14-1/2	--	--	--	--	--	X	--	--
3-5/8	X	--	X	X	--	--	X	--	14-3/4	--	--	--	--	--	X	--	--
3-11/16	X	--	X	X	--	--	X	--	15	--	--	--	--	--	X	--	--
3-3/4	X	--	X	X	--	X	X	--	15-1/4	--	--	--	--	--	X	--	--
3-13/16	X	--	X	X	--	--	X	--	15-1/2	--	--	--	--	--	X	--	--
3-7/8	X	--	X	X	--	--	X	--	15-3/4	--	--	--	--	--	X	--	--
3-15/16	X	--	X	X	--	--	X	X	16	--	--	--	--	--	X	--	--
4	X	--	X	X	--	X	X	X									
4-1/16	X	--	X	X	--	--	X	X									
4-1/8	X	--	X	X	--	--	X	--									
4-3/16	X	--	X	X	--	--	X	--									
4-1/4	X	--	X	X	--	X	X	--									
4-5/16	X	--	X	X	--	--	X	--									
4-3/8	X	--	X	X	--	X	X	--									
4-7/16	X	--	X	X	--	--	X	X									
4-1/2	X	--	X	X	--	X	X	X									
4-9/16	X	--	X	X	--	--	X	X									

* Larger sizes available on special order.

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TABLE IA. *Clamp range.*

Type A			Type B			Type C			Type D		
Nominal clamp size	A-diameter		Nominal clamp size	A-diameter		Nominal clamp size	A-diameter		Nominal clamp size	A-diameter	
	Max.	Min.		Max.	Min.		Max.	Min.		Max.	Min.
[illegible]	[illegible]	[illegible]	1/2	0.58	0.44	15/16	0.94	0.62	5/8	0.72	0.53
[illegible]	[illegible]	[illegible]	9/16	0.64	0.48	1	1.00	0.69	11/16	0.78	0.59
9/16	[illegible]	[illegible]	5/8	0.70	0.55	1-1/16	1.06	0.75	3/4	0.94	0.66
5/8	0.69	0.49	11/16	0.77	0.61	1-1/8	1.12	0.81	13/16	0.91	0.72
11/16	0.75	0.64	3/4	0.83	0.67	1-3/16	1.19	0.88	7/8	0.97	0.78
3/4	[illegible]	0.69	13/16	0.89	0.73	1-1/4	1.25	0.94	15/16	1.03	0.84
13/16	[illegible]	[illegible]	7/8	0.95	0.80	1-5/16	1.31	1.00	1	1.09	0.91
7/8	0.94	0.81	15/16	1.02	0.86	1-3/8	1.38	1.06	1-1/16	1.16	0.97
15/16	1.00	0.86	1-1/32	1.11	0.93	1-7/16	1.44	1.12	1-1/8	1.22	1.03
1	1.06	0.94	1-1/16	1.14	0.98	1-1/2	1.50	1.19	1-3/16	1.28	1.06
1-1/16	1.12	0.95	1-1/8	1.20	1.02	1-9/16	1.56	1.25	1-1/4	1.34	1.12
1-1/8	1.19	[illegible]	1-3/16	1.27	1.08	1-5/8	1.62	1.32	1-5/16	1.41	[illegible]
1-3/16	1.25	[illegible]	1-1/4	1.33	1.14	1-11/16	1.69	1.38	1-3/8	1.47	1.25
1-1/4	1.31	1.16	1-5/16	1.39	1.20	1-3/4	1.75	1.44	1-7/16	1.53	1.31
1-5/16	1.38	1.19	1-3/8	1.45	1.27	1-13/16	1.81	1.50	1-1/2	1.59	1.38
1-3/8	1.44	1.25	1-7/16	1.52	1.33	1-7/8	1.88	1.56	1-9/16	1.66	1.44
1-7/16	1.50	1.31	1-1/2	1.58	1.39	1-15/16	1.94	1.62	1-5/8	1.72	1.50
1-1/2	1.56	1.34	1-9/16	1.64	1.45	2	2.00	1.69	1-11/16	1.75	1.56
1-9/16	1.62	1.44	1-5/8	1.70	1.52	2-1/16	2.06	1.69	1-3/4	1.84	1.62
1-5/8	1.69	1.50	1-11/16	1.77	1.58	2-1/8	2.12	1.75	1-13/16	1.91	1.69
1-11/16	1.75	1.56	1-3/4	1.83	1.64	2-3/16	2.19	1.81	1-7/8	1.97	1.75
1-3/4	1.81	1.62	1-13/16	1.89	1.70	2-1/4	2.25	1.88	1-15/16	2.03	1.81
1-13/16	1.88	1.69	1-7/8	1.95	1.77	2-5/16	2.31	1.94	2	2.09	1.88
1-7/8	1.94	1.75	1-15/16	2.02	1.83	2-3/8	2.38	2.00	2-1/16	2.16	1.94
1-15/16	2.00	1.81	2-1/32	2.11	1.92	2-7/16	2.44	2.06	2-1/8	2.22	2.00
2	2.06	1.88	2-1/8	2.20	2.02	2-1/2	2.50	2.12	2-3/16	2.28	2.06
2-1/16	2.12	1.94	2-3/16	2.27	2.08	2-9/16	2.56	2.19	2-1/4	2.34	2.12
2-1/8	2.19	2.00	2-9/32	2.36	2.17	2-5/8	2.62	2.25	2-5/16	2.41	2.19
2-3/16	2.25	2.06	2-3/8	2.48	2.27	2-11/16	2.69	2.31	2-3/8	2.47	2.22
2-1/4	2.31	2.12	2-1/2	2.61	2.39	2-3/4	2.75	2.38	2-7/16	2.53	2.28
2-5/16	2.38	2.19	2-5/8	2.75	2.52	2-13/16	2.81	2.44	2-1/2	2.59	2.34
2-3/8	2.44	2.25	2-3/4	2.88	2.64	2-7/8	2.88	2.50	2-9/16	2.66	2.41
2-7/16	2.50	2.31	2-7/8	3.00	2.77	2-15/16	2.94	2.56	2-5/8	2.72	2.47
2-1/2	2.56	2.34	3	3.12	2.89	3	3.00	2.62	2-11/16	2.78	2.53
2-9/16	2.62	2.41	3-1/8	3.25	3.02	3-1/16	3.06	2.69	2-3/4	2.84	2.59
2-5/8	2.69	2.47	3-1/4	3.38	3.14	3-1/8	3.12	2.75	2-13/16	2.91	2.66
2-11/16	2.75	2.53	3-3/8	3.50	3.27	3-1/4	3.25	2.88	2-7/8	2.97	2.72
2-3/4	2.81	2.59	3-9/16	3.81	3.42	3-5/16	3.31	2.94	2-15/16	3.03	2.78
2-13/16	2.88	2.66				3-3/8	3.38	3.00	3	3.09	2.84
2-7/8	2.94	2.72				3-7/16	3.44	3.06	3-1/16	3.16	2.91
2-15/16	3.00	2.78				3-1/2	3.50	3.12	3-1/8	3.22	2.97
3	3.06	2.84				3-9/16	3.56	3.19	3-3/16	3.28	3.03
3-1/16	3.12	2.91				3-5/8	3.62	3.25	3-1/4	3.34	3.09
3-1/8	3.19	2.97				3-11/16	3.69	3.31	3-5/16	3.41	3.16
3-3/16	3.25	3.03				3-3/4	3.75	3.38	3-3/8	3.47	3.22
3-1/4	3.31	3.09				3-13/16	3.81	3.44	3-7/16	3.53	3.28
3-5/16	3.38	3.16				3-7/8	3.88	3.50	3-1/2	3.59	3.34
3-3/8	3.44	3.22				3-15/16	3.94	3.56	3-9/16	3.66	3.34
3-7/16	3.50	3.25				4	4.00	3.62	3-5/8	3.72	3.41
3-1/2	3.56	3.31				4-1/16	4.06	3.69	3-11/16	3.78	3.47
3-9/16	3.62	3.38				4-1/8	4.12	3.75	3-3/4	3.84	3.53
3-5/8	3.69	3.44				4-3/16	4.19	3.81	3-13/16	3.91	3.59
3-11/16	3.75	3.50				4-1/4	4.25	3.88	3-7/8	3.97	3.66
3-3/4	3.81	3.52				4-5/16	4.31	3.94	3-15/16	4.03	3.72
3-13/16	3.88	3.62				4-3/8	4.38	4.00	4	4.09	3.78
3-7/8	3.94	3.69				4-7/16	4.44	4.06	4-1/16	4.16	3.84
3-15/16	4.00	3.75				4-1/2	4.50	4.12	4-1/8	4.22	3.91
4	4.06	[illegible]				4-9/16	4.56	4.19	4-3/16	4.28	3.97
4-1/16	4.12	3.84				4-5/8	4.62	4.25	4-1/4	4.34	4.03
4-1/8	4.19	3.94				4-11/16	4.69	4.31	4-5/16	4.41	4.09
4-3/16	4.25	4.00				4-3/4	4.75	4.38	4-3/8	4.47	4.16
4-1/4	4.31	4.06				4-13/16	4.81	4.44	4-7/16	4.53	4.22
4-5/16	4.38	4.12				4-7/8	4.88	4.50	4-1/2	4.59	4.28
4-3/8	4.44	4.19				4-15/16	4.94	4.56	4-9/16	4.66	4.34
4-7/16	4.50	4.25				5	5.00	4.62	4-5/8	4.72	4.41
4-1/2	4.56	4.31							4-11/16	4.78	4.47
4-9/16	4.62	4.38							4-3/4	4.84	4.53
4-5/8	4.69	4.44							4-13/16	4.91	4.59
4-11/16	4.75	4.50							4-7/8	4.97	4.66
4-3/4	4.81	4.56							4-15/16	5.03	4.72
4-13/16	4.88	4.62							5	5.09	4.78
4-7/8	4.94	4.69							5-1/16	5.16	4.84
4-15/16	5.00	[illegible]							5-1/8	5.22	4.91
									5-3/16	5.28	4.97
									5-[illegible]	5.34	5.03
									[illegible]	[illegible]	5.09
									[illegible]	[illegible]	[illegible]
									5-7/16	5.53	5.22
									5-1/2	5.59	5.28
									5-9/16	5.66	5.34
									[illegible]	5.72	5.41
									5-11/16	5.78	5.47
									5-[illegible]	5.84	[illegible]
									5-13/16	[illegible]	5.59
									[illegible]	[illegible]	5.66
									[illegible]	[illegible]	5.72
									6	[illegible]	5.78

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TABLE IA. *Clamp range. (cont'd.)*

Type E			Type F			Type G			Type H		
Nominal clamp size	A-diameter		Nominal clamp size	A-diameter		Nominal clamp size	A-diameter		Nominal clamp size	A-diameter	
	Max.	Min.		Max.	Min.		Max.	Min.		Max.	Min.
3/8	.380	.370	25/32	0.78	0.44	1-1/2	1.500	.750	3/8	--	--
7/16	.442	.432	29/32	0.91	0.50	2-3/8	2.375	.750	7/16	--	--
1/2	.510	.490	1	1.00	0.50	4-5/8	4.625	2.000	1/2	--	--
9/16	.573	.551	1-1/16	1.06	0.56	6-1/2	6.500	4.000	9/16	--	--
5/8	.640	.610	1-1/4	1.25	0.69				5/8	--	--
11/16	.703	.671	1-3/8	1.38	0.50				11/16	--	--
3/4	.770	.730	1-1/2	1.50	0.81				3/4	--	--
13/16	.832	.792	1-3/4	1.75	0.81				13/16	--	--
7/8	.900	.850	1-15/16	1.94	1.25				7/8	--	--
15/16	.968	.906	2	2.00	1.06				15/16	--	--
1	1.031	.969	2-1/4	2.25	1.31				1	--	--
1-1/16	1.090	1.034	2-1/4	2.25	1.44				1-1/16	--	--
1-1/8	1.150	1.100	2-1/2	2.50	1.56				1-1/8	--	--
1-3/16	1.218	1.095	2-3/4	2.75	1.81				1-3/16	--	--
1-1/4	1.280	1.219	2-15/16	2.94	2.12				1-1/4	--	--
1-5/16	1.344	1.281	3	3.00	2.06				1-5/16	--	--
1-3/8	1.406	1.344	3-1/4	3.25	2.31				1-3/8	--	--
1-1/2	1.531	1.469	3-1/2	3.50	2.56				1-7/16	--	--
1-5/8	1.672	1.578	3-3/4	3.75	2.81				1-1/2	--	--
1-3/4	1.797	1.703	3-3/4	3.75	2.56				1-9/16	--	--
1-7/8	1.937	1.812	4	4.00	3.06				1-5/8	--	--
1-15/16	2.000	1.875	4-1/4	4.25	3.31				1-11/16	--	--
2	2.061	1.939	4-3/8	4.38	3.19				1-3/4	--	--
2-1/8	2.187	2.062	4-1/2	4.50	3.56				1-13/16	--	--
2-3/16	2.250	2.187	4-3/4	4.75	3.81				1-7/8	--	--
2-3/8	2.437	2.312	4-7/8	4.88	3.63				1-15/16	--	--
2-1/2	2.561	2.439	5	5.00	4.06				2	--	--
2-5/8	2.688	2.562	5-1/4	5.25	4.31				2-1/8	--	--
			5-1/2	5.50	4.25				2-3/16	--	--
			5-1/2	5.50	4.62				2-5/32	--	--
			5-3/4	5.75	4.87				2-1/4	--	--
			6	6.00	4.75				2-5/16	--	--
			6	6.00	5.12				2-3/8	--	--
			6-1/4	6.25	5.38				2-1/2	--	--
			6-1/2	6.50	5.56				2-9/16	--	--
			6-3/4	6.75	5.88				2-5/8	--	--
			7	7.00	6.12				2-13/16	--	--
			7-1/4	7.25	6.38				2-3/4	--	--
			7-1/2	7.50	6.62				2-7/8	--	--
			7-3/4	7.75	6.88				3	--	--
			8	8.00	7.12				3-1/16	--	--
			9	9.00	8.12				3-1/8	--	--
			10	10.00	9.12				3-5/16	--	--
			11	11.00	10.12				3-3/8	--	--
			12	12.00	11.12				3-1/2	--	--
			13	13.00	12.12				3-9/16	--	--
			14	14.00	13.12				3-15/16	--	--
			15	15.00	14.12				4	--	--
			16	16.00	15.12				4-1/16	--	--
									4-7/16	--	--
									4-1/2	--	--
									4-9/16	--	--
									5	--	--
									6	--	--
										--	--
										--	--

3.3 Protective coatings.

3.3.1 *Cadmium plating.* When cadmium plating is specified (see 6.2), carbon or spring steel clamps and components shall be plated in accordance with QQ-P-416, type II, class 3.

3.3.2 *Zinc plating.* When zinc plating is specified (see 6.2), carbon or spring steel clamps and components shall be plated in accordance with QQ-Z-325, type II, class 3.

3.3.3 *Passivation.* When a passivation treatment for clamps made of corrosion-resisting steel is specified (see 6.2), the clamps shall be treated in accordance with QQ-P-35.

3.3.4 *Embrittlement relief.* Cadmium or zinc plated steel clamps having a hardness of Rockwell C40 or higher, shall be subjected to an embrittlement relief treatment (baking) in accordance with the applicable plating specification.

3.4 Performance and product characteristics.

3.4.1 *Free torque.* The free torque, in the uninstalled condition, to turn the screw on the adjustable type clamps, up to and including 1-1/4 inch diameter, shall not exceed 8 inch-pounds. The free torque of screws for adjustable type clamps larger than 1-1/4 inch diameter shall not exceed 3 inch-pounds.

3.4.2 *Durability.* Type F clamps shall withstand an applied screw torque of 50 inch-pounds for carbon steel screws and 60 inch-pounds for corrosion-resisting steel screws, without evidence of deformation or excessive wear on the screw threads or slots in the band.

3.4.3 *Load vs wrench torque.* The clamping loads for types A,B,C, and D clamps shall not be less than 200 pounds for 20 inch-pounds of torque. The clamping load for type F clamps shall not be less than 400 pounds for 40 inch-pounds of torque. Type G clamps shall not be less than 400 pounds for 20 inch-pounds of torque (see 6.2).

3.4.4 *Release torque.* The average torque required to release the adjustable type clamps shall not be less than 6 inch-pounds.

3.4.5 *Hydraulic pressure.* Clamps shall be capable of withstanding the pressures for the applicable sizes as specified in table II.

3.4.6 *Band ductility.* All types of clamp bands, except type E, shall be capable of withstanding a 180° bend around a 3/16 inch diameter arbor and restraightening, without any evidence of fracture, breakage, or cracks. With type E clamps gripped as shown in figure 4, the free tang of the clamp shall be capable of passing around the stationary tang without evidence of breaks, cracks, or fractures.

3.4.7 *Vibration resistance.* Clamps shall withstand the vibration resistance test of 4.5.9. There shall be no evidence of loosening or indications of failure and the clamps shall seal with 20 inch-pound torque (see 6.2).

3.4.8 *Damage to hose.* There shall be no damage to the outer covering of the hose used in the applicable tests of 4.5 due to imperfections of the hose clamp.

3.5 Identification marking. Unless otherwise specified (see 6.2), when size permits, each clamp shall be marked in a plain and permanent manner, with the size and the manufacturer's trademark or name.

3.6 Workmanship. Clamps shall be free from burrs, sharp edges, cracks, malformation, grit, dirt, and abrasives.

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TABLE II. *Hydraulic test pressures*

Hose inside diameter (inches)	Types A,B,C, and D 20 in. lbs. torque	Type E	Type H	Type G 20 in. lbs. torque	Type F 40 in. lbs. torque
	P.s.i.	P.s.i.	P.s.i.	P.s.i.	P.s.i.
5/16 or less	600	50	600	600	600
3/8 to 1/2	470	50	600	600	600
1/2	350	50	600	600	600
5/8	280	50	600	350	350
3/4 to 1	235	50	500	350	350
1	175	50	500	350	350
1-1/8 to 1-1/4	150	50	500	300	300
1-3/8 to 1-1/2	150	50	400	300	300
1-3/4	125	50	300	250	250
2	125	50	275	250	250
2-1/4 to 4	100	50	200	200	200
4-1/4 to 6	75	50	150	100	100

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Lot formation. A lot shall consist of all clamps of the same material, type and size, manufactured under the same conditions. Formation of lots shall be in accordance with MIL-STD-105.

4.3 Sampling.

4.3.1 *Sampling for visual and dimensional examination.* A random sample of clamps shall be taken from each lot in accordance with MIL-STD-105, inspection level II. The acceptable quality level (AQL) shall be as specified in table III.

4.3.2 *Sampling for tests.* A random sample of clamps shall be taken from each lot in accordance with MIL-STD-105, inspection level S-2. The AQL shall be 1.5 percent.

4.3.3 *Sampling for protective coatings.* Sampling for tests of protective coatings shall be in accordance with the applicable specifications referenced in 3.3.1 through 3.3.3.

4.3.4 *Sampling for packaging and packing.* Sampling for inspection of packaging and packing shall be in accordance with MIL-H-775.

TABLE III. *Classification of defects*

Categories	Defects	Inspection method
Critical	None defined	
Major:	AQL = 1.5 percent	
101	Dimensions	Measure
102	Sharp edges	Visual and tactile
103	Cracks, burrs	Visual
104	Incomplete assembly	Visual
Minor:	AQL = 2.5 percent	
201	Protective coating (when specified) missing	Visual
202	Identification marking	Visual
203	Malformations	Visual
204	Workmanship	Visual

4.5 Tests.

4.5.1 *Test procedures.* Each clamp in the sample of 4.3.2 shall be subjected to the

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tests of 4.5.3 through 4.5.9, except that tests of 4.5.4—"Durability", 4.5.5—"Load vs wrench torque", 4.5.7—"Hydraulic pressure" and 4.5.9—"Vibration" for clamps 6 inches in diameter and larger will not be required.

4.5.2 *Test conditions.* Test samples shall not be worked, manipulated, or lubricated to increase the freeness with which the clamps operate, prior to conducting the tests specified herein.

4.5.3 *Free torque.* The wrench torque required to turn the adjusting screw of the clamp in the uninstalled condition shall be measured and recorded over the complete range of its adjustment to determine conformance to 3.4.1.

4.5.4 *Durability.* Type F clamps shall be installed and tightened once on a steel test mandrel, similar to figure 2, to an applied screw torque of 50 inch-pounds for carbon steel screws and 60 inch-pounds for corrosion-resistant screws, to determine conformance to 3.4.2.

4.5.5 *Clamping load versus wrench torque.* When a load vs wrench torque test is specified (see 6.2), types A,B,C,D,F, and G clamps shall be installed on a split hose, which in turn is installed on the spacer blocks of the hose clamp tester (see figure 3). The stretch indicator shall be set at zero when the spacer blocks are together.

4.5.5.1 With the clamp contracting the full periphery of the hose and with 0 to 3 inch-pounds torque on the clamp adjusting screw, sufficient hydraulic pressure shall be applied to the tester to show 0.150 to 0.200 inch on the deflection indicator.

4.5.5.2 The shut-off valve between the pump and the piston shall then be closed. A torque of 20 inch-pounds shall then be applied to the adjusting screw of types A,B, C,D, and G clamps. A torque of 40 inch-pounds shall be applied to the type F clamp adjusting screw. Clamping loads shall conform to 3.4.3.

4.5.5.3 If necessary to provide take-up (for maintenance of minimum distance between spacer block), the shut-off valve may be opened and sufficient hydraulic fluid pumped into the cylinder of the tester so that the stretch indicator will be within 0.015 and 0.150 inch before increasing the torque to the next valve, provided the load of the next wrench torque is neither equal to nor exceeded.

4.5.6 *Release torque.* Types A,B,C,D, and G clamps shall be installed on the test hose and mandrel as shown in figure 2 and torqued to 20 inch-pounds and type F clamps shall be installed similarly and torqued to 25 inch-pounds and released five times, to determine conformance with 3.4.4.

4.5.7 *Hydraulic pressures.* Two clamps of the same type and size shall be installed over a 3-inch length of clean, dry hose conforming to MIL-H-6000. The hose shall be internally coated with a thin film of oil conforming to grade 1100 of MIL-L-6082. The hose shall be installed over a clean test mandrel (see figure 2) and the adjustable type clamps torqued as specified in table II. Types E and H clamps shall be installed as recommended by the manufacturer.

4.5.7.1 Types A,B,C,D,E,F,G, and H clamps shall meet their respective pressures specified in table II. Evidence of oil leakage between the test hose and the mandrel shall be a basis for rejection of the clamp.

4.5.8 *Band ductility.* The bands of types A,B,C,D,F,G, and H clamps shall be subject to a 180° cold bend around a 3/16 inch diameter mandrel and then restraightened to determine conformance to 3.4.6.

4.5.8.1 Type E clamps shall be gripped in a vise as shown in figure 4. The clamp shall be expanded by moving the free tang of the clamp to a point where the free tang will pass around the stationary tang to determine conformance to 3.4.6.

4.5.9 *Vibration resistance.* When a vibra-

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tion resistance test is specified (see 6.2), types A,B,C,D,F, and G clamps shall be tightened to 20 inch-pounds of torque. The position of the adjusting devices with relation to their respective saddles shall be marked. Types E and H clamps shall be installed as recommended by the manufacturer.

4.5.9.1 The hose assembly shall then be placed in a vibration testing machine as shown in figure 5. One end of the hose assembly shall be vibrated at 1/4-inch double amplitude at 1800 cycles per minute for 23 hours. During this test, oil conforming to MIL-L-6082, grade 1100 shall be pumped into the assembly at a pressure of 40 to 50 pounds per square inch (p.s.i.).

4.5.9.2 After completion of the vibration period, the clamps shall be examined to determine conformance to 3.4.7. If there is leakage and the adjustable type clamps show a torque of less than 20 inch-pounds due to set on the hose, the adjustable type clamps shall be tightened to 20 inch-pounds torque and again examined for leakage. There shall be no leakage when sealed at 20 inch-pounds torque.

4.5.10 *Damage to hose.* The hose, figure 2, used in the tests above, as applicable, shall be examined for resultant damage to determine compliance with 3.4.8.

4.5.11 *Protective coating test.* Test of protective coating shall be in accordance with the applicable specifications referenced in 3.3.1 through 3.3.3.

4.5.12 *Inspection of packaging, packing, and marking.* Inspection of packaging, packing, and marking shall be in accordance with MIL-H-775.

5. PREPARATION FOR DELIVERY

5.1 **Preservation, packaging, packing, and marking.** Preservation, packaging, packing, and marking of the clamps shall be in accordance with MIL-H-775.

5.2 **Level.** Levels of packaging and packing shall be as specified by the procuring activity (see 6.2).

6. NOTES

6.1 **Intended use.** Clamps covered by this specification are used for holding low-pressure, flexible ducts, or hose on to a male fitting.

6.2 **Ordering data.** Purchasers should exercise any desired options offered herein and procurement documents should specify the following:

(*a*) Title, number, and date of this specification.
(*b*) Type and size of clamp.
(*c*) Material.
(*d*) Protective coating, when required (see 3.3).
(*e*) Level of packaging and packing (see 5.2).
(*f*) Identification marking, when required (see 3.5).
(*g*) Vibration test, when required (see 4.5.9).
(*h*) Load versus wrench torque test, when required (see 4.5.5).

6.3 **Military procurement.** Items procured under this specification for military use are to be limited to the variety shown on the applicable military standards. Personnel of the military departments are requested to refer to these documents for guidance.

6.4 Cross reference data between WW-C-440a and WW-C-440b is as follows:

WW-C-440a	*WW-C-440b*
Type I	Type H
Type II—Style 1	Type F
Type II—Style 2	Types B and D
Type III	Type C
Type IV	Type E
Type V	Type A
None	Type G

6.5 **Patent notice.** The slotted type G clamp of figure 1 is covered by Patent Number 2945277, dated July 19, 1960. The Gov-

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ernment does not have a royalty free license under this patent.

6.6 **Transportation description.** The transportation descriptions and minimum weights applicable to this commodity are:

Rail:

Hose clamps, steel—carload minimum weight 30,000 pounds.

Motor:

Hose clamps, steel—truckload minimum weight 30,000 pounds, subject to Rule 115, National Motor Freight Classification.

Rail:

Hose clamps, metal, not otherwise indexed by name.
Carload minimum weight 30,000 pounds.

Motor:

Hose clamps, metal, not otherwise indexed by name.
Truckload minimum weight 30,000 pounds, subject to Rule 115, National Motor Freight Classification.

CUSTODIANS:

Army—WC
Navy—YD
Air Force—69

***Review activities*:**

Army—GL, MI, MO, WC
Navy—YD, SH
Air Force—69
DSA—CS

User activities:

Army—MU
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***Preparing activity*:**

Army—WC

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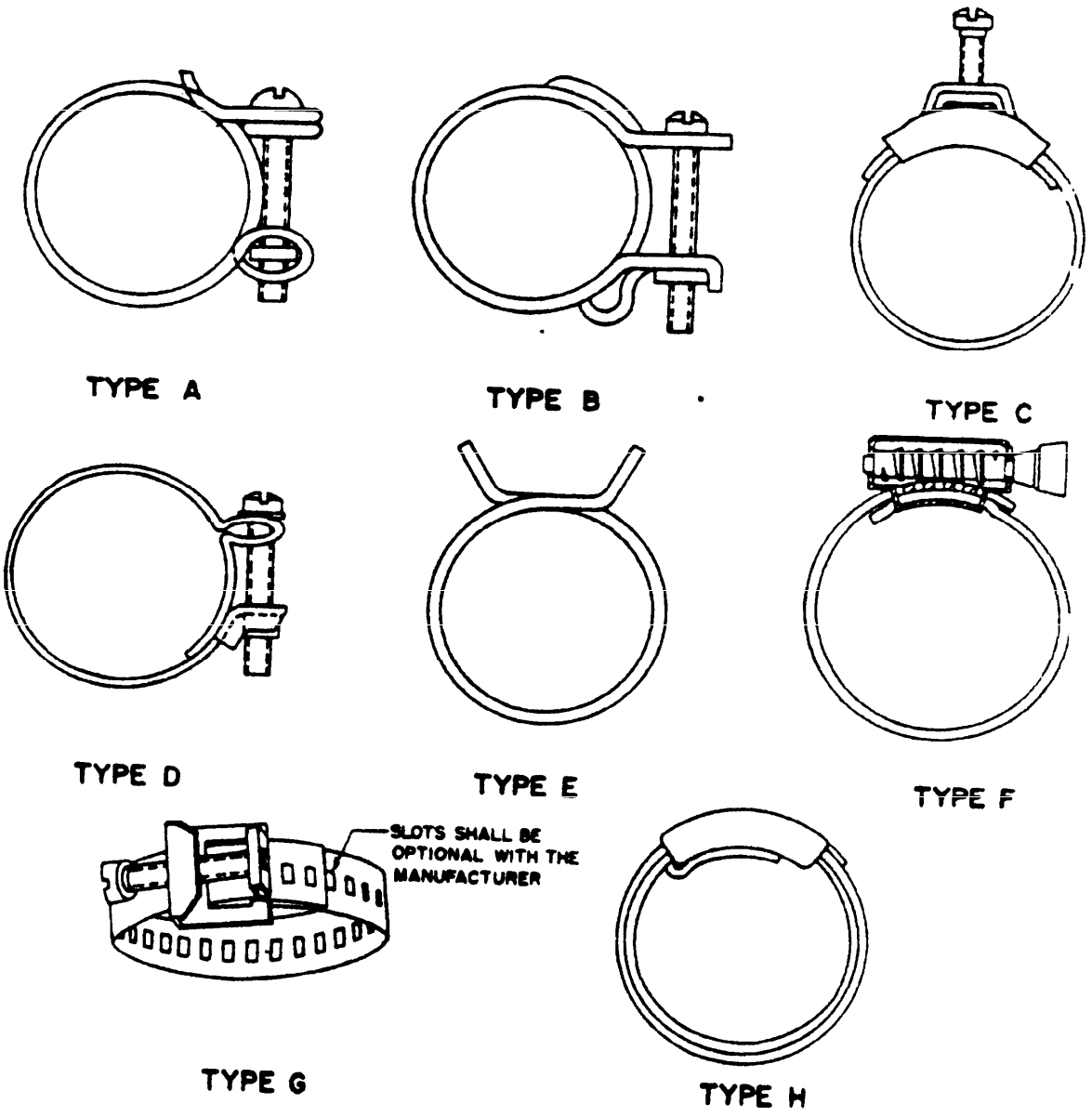


FIGURE 1. Clamp, hose

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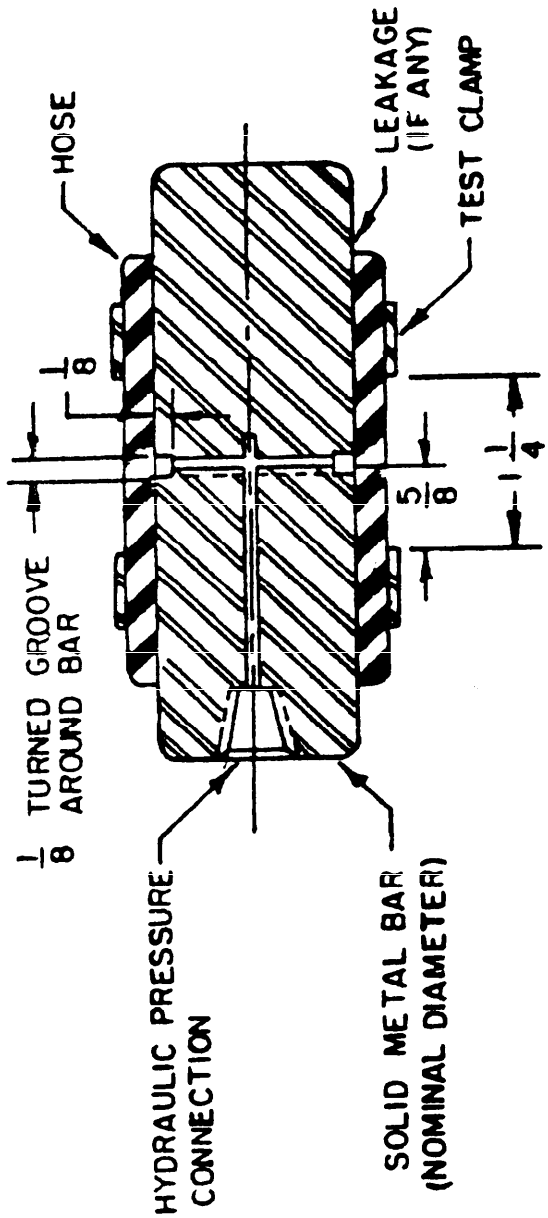


FIGURE 2. Hydraulic pressure test

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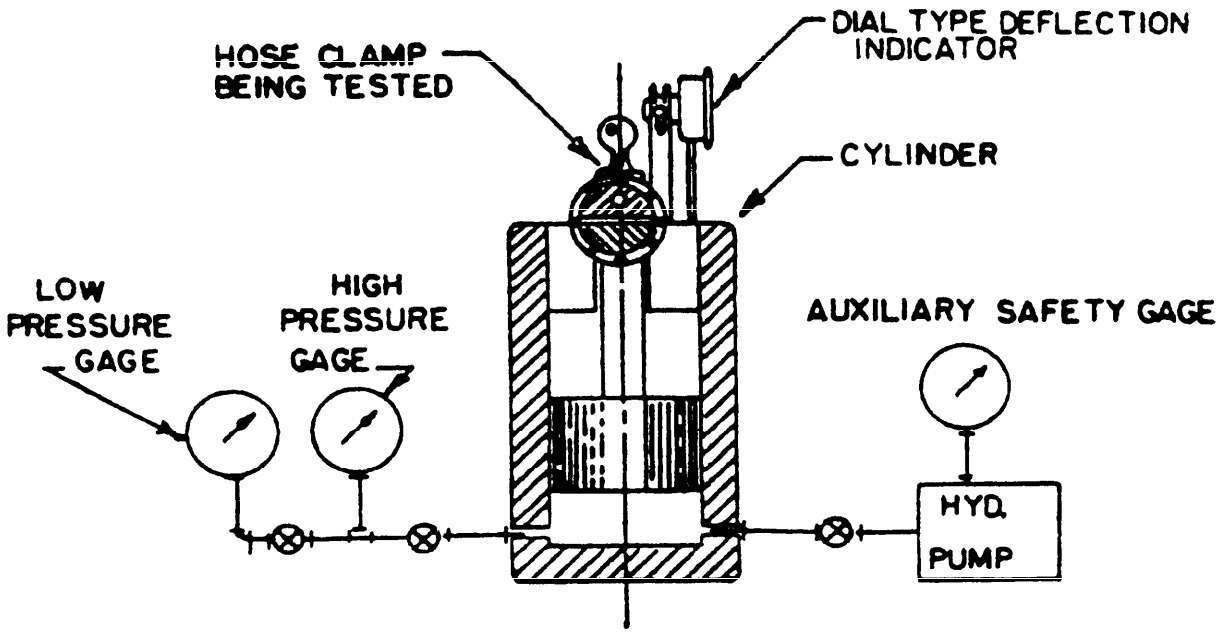


FIGURE 3. Load versus torque test

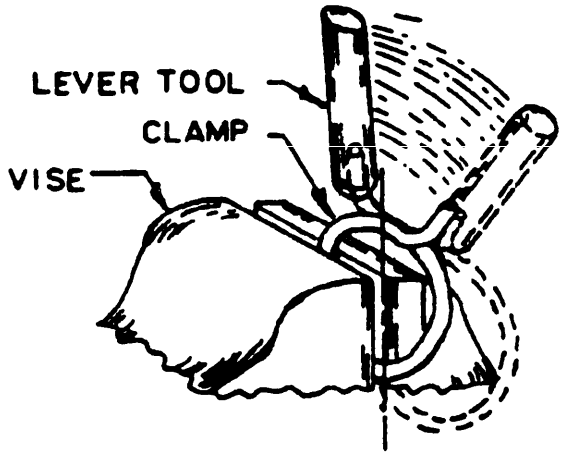


FIGURE 4. Ductility test—spring type clamp

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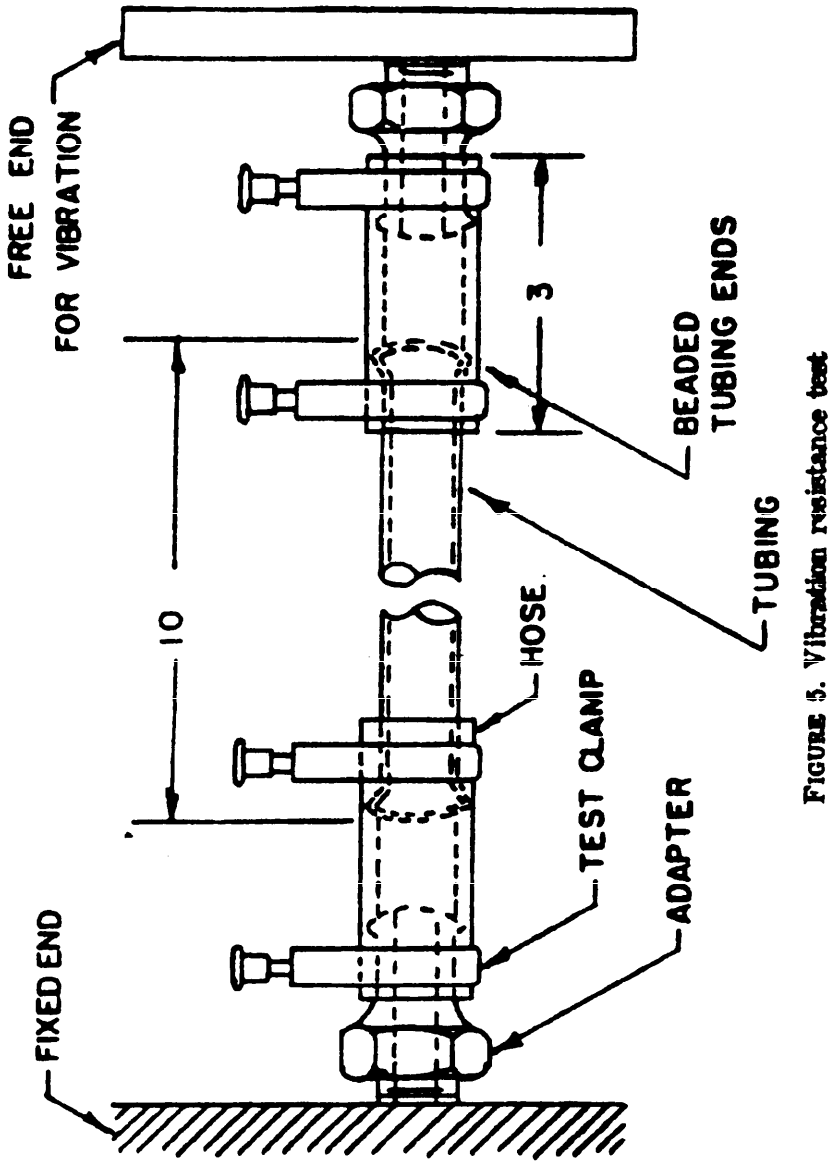


FIGURE 5. Vibration resistance test

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